Architectural Fidelity

Empirical Observations on Data Structure & Memory Latency & Impact on Performance

SHEIKH MOHSIN

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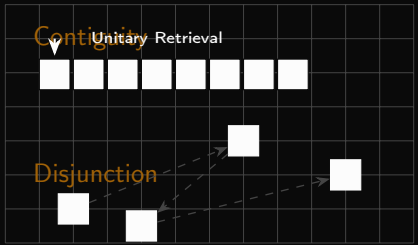
The Spatial Premise



- Modern CPUs operate on a **spatial premise**.
- Contiguous layouts leverage hardware prefetching.
- Pointer-chasing induces serial stalls.
- Memory latency remains the primary bottleneck of 21st-century systems.

The Scale of the Project

16
Primitive Data Structures

94
Datapoints

7.0B
Memory Elements Processed

Methodology

Each datapoint represents the median of up to 1,000 statistical samples collected via internal hardware performance counters, filtered for a Coefficient of Variation (CV) $< 15\%$.

Taxonomy: The 16 Contenders

Contiguous
Predictable Layout

- Array / Vector
- Circular Buffer
- Binary Heap
- AoS vs SoA

Link-Based
Stochastic Locality

- Linked List
- Skip List
- Slab List

Trees & Hybrids
Logic-Over-Layout

- BST / RBTree
- BTree / vEBTree
- HashMap / Trie
- Deque

Each structure was implemented in C11 and measured across four decades of scale (10^3 to 10^7).

Methodology I: Architectural Determinism

System Isolation
Eliminating OS-Level Variance

- **CPU Pinning**: Benchmarks pinned to an isolated core to prevent context-switch overhead.
- **Performance Governor**: Frequency locked at max clock to eliminate P-state ramp-up noise.
- **ASLR Disabled**: Ensuring deterministic virtual-to-physical address mapping.

Hardware State
Consistency Across Runs

- **Cold/Warm Control**: 3 leading warmup passes to prime branch predictors and L1/L2 caches.
- **C-State Suppression**: Throttling deep-sleep states to maintain constant pipeline readiness.

Methodology II: Discrete Instrumentation

```
start_counters(fds);

for(int i=0; i < N; i++) {
    sink += traversal(i);
}

stop_counters(fds, &sample);
```

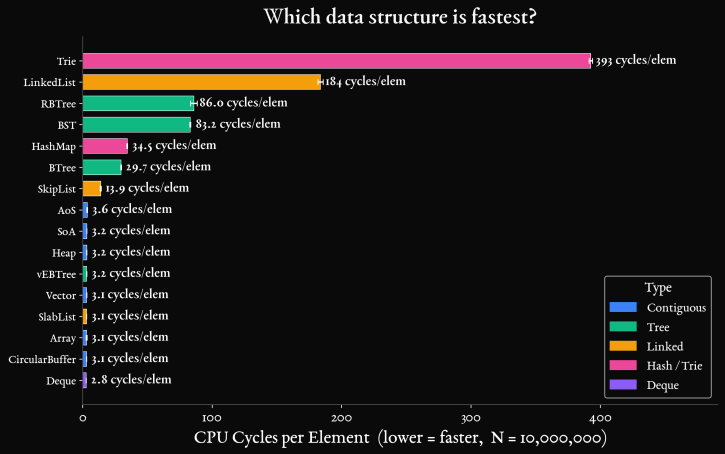
→ L1/L3 Misses

→ Cycles/IPC

Direct Interface
The **perf_event** Subsystem

- **Zero-Overhead Measurement**: Internal syscalls wrap the precise observation window.
- **Median Filtering**: Capturing 1,000 samples per structure. We report the **Median** to discard preemption-induced outliers.
- **Noise Guard**: Automatic discard of results with CV $> 15\%$.

Empirical Observations: 16 Primitive Structures



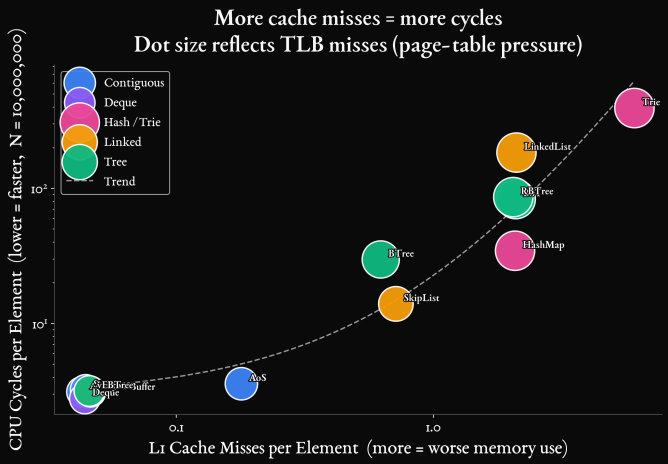
The Latency Gap

A **57**× divergence exists between the fastest and slowest structures.

Contiguous systems (<4 cycles) leverage the full pipeline width, while Linked systems (>170 cycles) remain stalled on memory signals.

Measured in CPU cycles per element at $N = 10,000,000$. Sorted by architectural efficiency.

The Causality of Cache Exhaustion



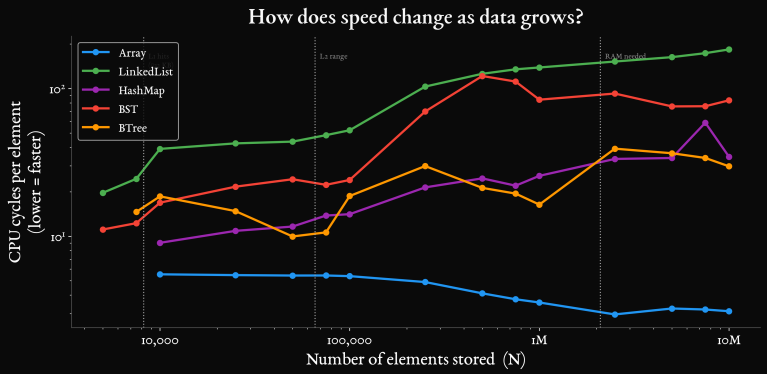
The Linear Drift

The correlation between L1 cache-invalidation and net latency is near-perfect ($R^2 \approx 0.99$).

Observation:

If representation fails the cache, logic cannot recover the speed.

Working-Set Dynamics & Performance Inversion



Log-log scale. Divergence occurs at cache-boundary saturation.

A Proposed Hierarchy of Data Representation

The Ideal
Arrays & Vectors
Optimized for the physical machine.

The Hybrid
BTrees
Clustered nodes to amortize latency.

The Suboptimal
Linked Lists
Architecturally inefficient at scale.

Computational complexity is secondary to spatial complexity on modern hardware.

Scientific Disclosure: Limitations

Architectural Specificity

Hardware-Dependent Results

- **Microarchitecture**: Results are specific to the Intel/AMD cache hierarchy used during testing.
- **Single Operation**: Benchmarks isolate **Traversal** performance; read/write/delete trade-offs are not explored here.

Methodological Scope

Trade-offs in Modeling

- **Allocation Bias**: Measurements exclude **malloc** overhead; real-world fragmentation impacts are simplified.
- **Stationary Workloads**: Does not model dynamic churn or cache-pollution from interleaved tasks.

Synthesis

I. Memory is no longer a passive storage tier, but a performance engine.
II. Representation dictates execution speed more than algorithmics.
III. Contiguity is a prerequisite for high-fidelity systems.

Full project at github.com/SheikhMohsin9311/Data-Structure-Performance-Analysis